

Current and Resistance

6.1. **Electric current.** In an isolated metallic conductor, the free electrons present in it are in random motion like the molecules of a gas. So the net rate at which electrons pass through any hypothetical plane is zero. If the ends of the conductor are connected to a battery, an electric field E will be set up at every point within the conductor. This field acts on the electrons and gives them a resultant motion in the direction $-E$. If a charge q passes through any cross-section of the conductor in time t , then the current i is defined by

$$i = q/t$$

When q is in coulomb and t in second, then i is in ampere. If the rate of flow of charge with time is not constant, the current varies with time. Then at any instant t current is defined as

$$i = dq/dt$$

So **e.** Current is a *scalar* quantity. Conventionally, the direction of electric current is taken along the direction of motion of positive charges. When the current is caused by the motion of electrons, the direction of current is opposite to the direction of electron flow.

y. Current is a macroscopic quantity. The related microscopic quantity is the *current density* J .

T **It**

It is a vector quantity and is characteristic of the given point inside a conductor. If the current is flowing uniformly across a conductor of cross-sectional area A , the *magnitude* of the current density for all points on that cross section is given by

$$J = \frac{q/t}{A} = \frac{i}{A}$$

The unit of current density is A/m^2 .

Current i is related to the current density $\mathbf{J}$ by

$$i = \int \mathbf{J} \cdot d\mathbf{S}$$

where $d\mathbf{S}$ is an element of area and the integral is taken over any surface cutting across the conductor.

From this equation we may define the current i as the flux of the current density vector $\mathbf{J}$ through a given area.

6.2.

Consider a wire of length L and area of cross section A in which there is an electric field $\mathbf{E}$ towards the left (Fig. 6.1). Let n be the number of free electrons per metre³. The free electrons move towards the right with a constant drift velocity v_d .

Number of conduction electrons in the wire = $n(AL)$.

Let e be the charge on each electron.

Magnitude of charge q , passing out of the wire in time t is

$$q = (nAL)e$$

where t is given by

$$t = L/v_d$$

The current in the wire is

$$i = \frac{q}{t} = \frac{nALe}{L/v_d} = nAev_d$$

But i/A is the current density J . Therefore

$$J = nev_d$$

In vector form,

$$\mathbf{J} = (ne)\mathbf{v}_d \quad \dots(1)$$

(ne) is the volume charge density ρ of the current carriers. For positive carriers ne is positive and Eq. (1) predicts that $\mathbf{J}$ and $\mathbf{v}_d$ point in the same direction. For electrons $\mathbf{J}$ and $\mathbf{v}_d$ point in opposite directions.

6.3. Equation of Continuity

Consider a closed surface S enclosing a region of volume V . Let the surface enclose some charge within it. The current i flowing out through the surface S is given by

$$i = \oint_S \mathbf{J} \cdot d\mathbf{S} \quad \dots(i)$$

where $\mathbf{J}$ is the current-density vector over a cross-section $d\mathbf{S}$.

Let ρ be the charge density (charge residing per unit volume) of the

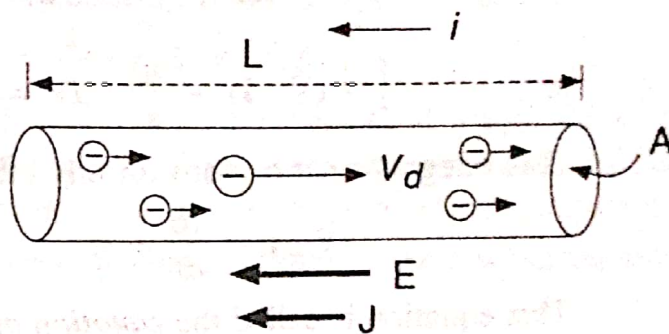


Fig. 6.1

charge enclosed within the surface at any instant. Then the total charge enclosed at any instant is given by

$$q = \int_V \rho dV$$

The current i is the rate of decrease of charge inside the volume, so that

$$i = - \frac{dq}{dt} = - \frac{d}{dt} \int_V \rho dV = - \int_V \frac{d\rho}{dt} dV \quad \dots(ii)$$

According to Gauss divergence theorem, $\oint_S \mathbf{J} \cdot d\mathbf{S} = \int_V (\nabla \cdot \mathbf{J}) dV$

$$\therefore \oint_S \mathbf{J} \cdot d\mathbf{S} = \int_V (\nabla \cdot \mathbf{J}) dV = - \int_V \frac{d\rho}{dt} dV$$

$$\text{or} \quad \int_V \left\{ (\nabla \cdot \mathbf{J}) + \frac{d\rho}{dt} \right\} dV = 0$$

This integral must be zero for any arbitrary volume. So

$$\nabla \cdot \mathbf{J} + \frac{d\rho}{dt} = 0 \quad \dots(iii)$$

This equation is called the *equation of continuity*. It simply expresses the fact that the charge is conserved.

In most practical situations of interest, the charge density is constant in time at each point. This means $d\rho/dt$ is zero at each point. This is called a *steady state*. Hence for *steady currents*

$$\nabla \cdot \mathbf{J} = 0 \quad \dots(iv)$$

6.4. Ohm's Law

It is found experimentally that in a metallic conductor at constant temperature, the current density $\mathbf{J}$ is linearly proportional to the electric field $\mathbf{E}$. Thus

$$\mathbf{J} = \sigma \mathbf{E} \quad \dots(1)$$

The equation $\mathbf{J} = \sigma \mathbf{E}$ is general vector statement of Ohm's law. Here σ is a constant called the *conductivity* of the conductor. The unit of conductivity is mho/metre. The reciprocal of the conductivity is called the *resistivity* ρ . In terms of ρ Eq. (1) can be written as

$$\mathbf{E} = \rho \mathbf{J} \quad \dots(2)$$

In Fig. 6.2 the electric field along the rod is in the direction MN and its value is $E = V/L$ everywhere. Here V is the total potential drop from M to N . Thus

$$J = \sigma (V/L)$$

The total current =

$$i = JA = \sigma VA/L$$

Here A is the cross-sectional area of the rod. This leads to

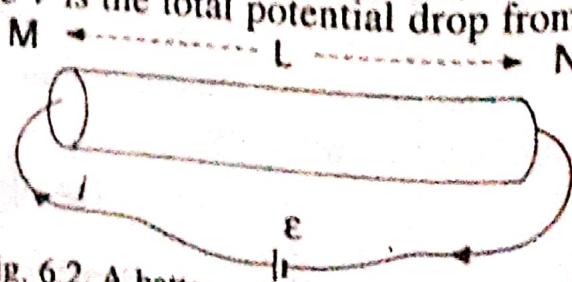


Fig. 6.2. A battery of emf ϵ can maintain an electric field within a conducting rod.

$$\frac{V}{i} = \frac{L}{\sigma A} = \frac{\rho L}{A} \quad \dots(3)$$

The ratio V/i is called the *resistance* R of the rod. $R = \rho L/A$.

Eq. (3) becomes $V = iR$

This is known as Ohm's law.

Experiments show that generally ρ and therefore R increase with temperature according to

$$R_t = R_0 (1 + \alpha t)$$

where α is called the *temperature coefficient of resistivity*, and t is the temperature in degrees centigrade.

The *power* dissipated in a resistance R carrying a current i is

$$P = i^2 R = iV = V^2/R$$

e. A copper wire of diameter 0.5mm and length 20 m is connected across a battery of emf 1.5 V and internal resistance 1.25Ω . Calculate the current density in the wire and the drift velocity v_d , assuming one conduction electron per atom of copper. What is the heat dissipated per metre of the wire ?

Sol. The total resistance of the wire is

$$R = \frac{\rho L}{A} = \frac{(1.7 \times 10^{-8}) \times 20}{\pi (0.25 \times 10^{-3})^2} = 1.732\Omega$$

$$\text{Current} = i = \frac{V}{R_{\text{total}}} = \frac{1.5}{(1.732 + 1.25)} = 0.5 \text{ A.}$$

$$\text{Resistance of unit length of wire} = 1.732/20 = 0.0866 \Omega/\text{m}$$

$$\text{Power dissipated per metre} = i^2 R = (0.5)^2 (0.0866) = 0.0216 \text{ W/m}$$

The number of copper atoms per m^3 is

$$n = \frac{\text{Avogadro No.} \times \text{density}}{\text{atomic weight}} \\ = \frac{(6.025 \times 10^{26}) (8.9 \times 10^3)}{63.54} = 8.43 \times 10^{28}$$

$$\text{Current density, } J = \frac{i}{A} = \frac{0.5}{\pi (0.25 \times 10^{-3})^2} = 2.546 \times 10^6 \text{ A m}^{-2}$$

$$v_d = \frac{J}{ne} = \frac{2.546 \times 10^6}{(8.43 \times 10^{28}) (1.6 \times 10^{-19})} = 0.1888 \times 10^{-3} \text{ m s}^{-1}$$

6.5. Drude-Lorentz Theory of Electrical Conduction

The basic assumptions of Lorentz-Drude theory are :

1. There are a large number of free electrons in a metal. These electrons are free to move about the whole volume of the metal like the

molecules of a perfect gas in a container.

2. The free electrons make collisions from time to time with fixed positive ions in the lattice and also among themselves. We may neglect the collisions between electrons in comparison with the collisions between electrons and ion cores.

3. In the absence of electric field, the random motion of free electrons is equally probable in all directions so that current density vector is zero.

4. When an external electric field is applied, the electrons drift slowly with some average velocity, known as *average drift velocity*, in the direction opposite to that of electric field. The drift velocity of free electrons is superimposed over their random velocity. In Fig. 6.3, the continuous solid lines show a possible random path followed by an electron in the absence of an applied field. The dashed lines show the electron path in the presence of the electric field E . Note the steady drift in the direction of $-E$. The drift velocity is much smaller in magnitude than average random velocity of free electrons.

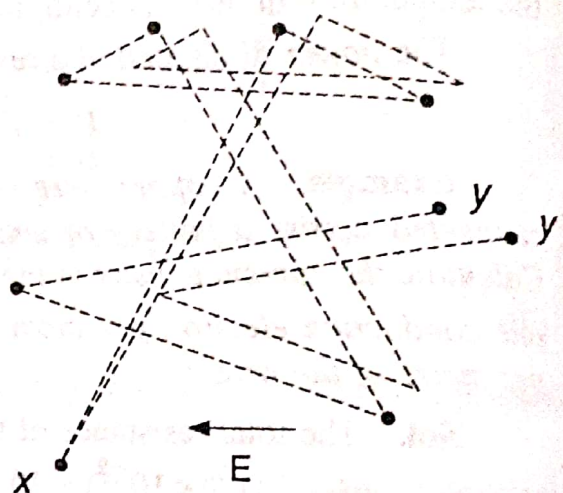


Fig. 6.3

5. The average distance traversed by a free electron between two successive collisions with the positive ions is called *mean free path* and is denoted by λ .

If an electron of mass m and charge e is placed in an electric field E , it experiences an acceleration a given by

$$a = eE/m.$$

Let τ be the average time-interval between two successive collisions of an electron with positive ions.

Increase in drift velocity in time $\tau = a\tau = eE\tau/m$

This drift velocity is reduced to zero at the next collision and again builds up to the value $eE\tau/m$ before suffering another collision and so on.

$$\therefore \left. \begin{array}{l} \text{the average drift velocity} \\ \text{through the conductor} \end{array} \right\} = v_d = \frac{eE\tau}{2m}$$

Let n be the number of conduction electrons per unit volume.

$$\text{Current density} = J = nev_d = \frac{ne^2E\tau}{2m}$$

But $J = \sigma E$ where σ is the electrical conductivity of the metal.

$$\therefore \sigma = \frac{J}{E} = \frac{ne^2\tau}{2m}$$

This is the expression for the electrical conductivity of a metal.

(Note. Some authors write the above equation without the factor $1/2$.)

$$\text{Resistivity} = \rho = 1/\sigma = 2m/ne^2\tau$$

Let λ be the mean free path and v the r.m.s. thermal velocity of electrons. Then $\tau = \lambda/v$

$$\text{Electrical conductivity, } \sigma = \frac{ne^2\lambda}{2mv}$$

The kinetic energy of a free electron, $\frac{1}{2}mv^2 = \frac{3}{2}kT$.

$$\therefore \sigma = \frac{ne^2\lambda v}{2mv^2} = \frac{ne^2\lambda v}{6kT}$$

Deduction of Ohm's law From Drude-Lorentz theory

From Drude Lorentz theory, $J = \sigma E = \frac{ne^2\lambda}{2mv} E$

At a particular temperature, λ and v are constant.

$$\therefore J \propto E$$

Let i be the current flowing through the conductor of length l and area of cross-section A and V be the potential difference across it. Then

$$J = \frac{i}{A} \text{ and } E = \frac{V}{l}$$

The above relation becomes

$$\frac{i}{A} \propto \frac{V}{l}$$

For a given conductor, l and A are constant at a given temperature.

$$\therefore i \propto V$$

i.e., at a given temperature, the current flowing through a conductor is proportional to the potential difference between its two ends, which is Ohm's law.

Variation of Electrical Conductivity with Temperature

The electrical conductivity of metals is given by

$$\sigma = \frac{ne^2\tau}{2m}$$

According to the kinetic theory of gases (here we are dealing with electron-gas) the time-interval τ between two collisions is inversely proportional to the square-root of the absolute temperature :

$$\tau \propto \frac{1}{v_{rms}} \propto \frac{1}{\sqrt{T}}.$$

Therefore, the conductivity σ must decrease with increase in temperature. Experimentally, however, σ is inversely proportional more nearly to the first power of the absolute temperature.

Wiedemann-Franz Law. By an analysis similar to that used for electrical conductivity, it may be shown that the thermal conductivity K is given by

$$K = \left(\frac{3nk^2\tau}{2m} \right) T,$$

where k is Boltzmann's constant and T is the absolute temperature.

$$\therefore \frac{K}{\sigma} = 3 \left(\frac{k}{e} \right)^2 T.$$

This relation is known as the *Wiedemann-Franz law*.

Statement. *The ratio of thermal conductivity to electrical conductivity is a constant for all metals and is proportional to the absolute temperature.*

6.6. Kirchhoff's Laws

Law 1. *In any network of conductors, the algebraic sum of the currents meeting at any point is zero. i.e., $\sum I = 0$*

In the above equation, current flowing into a junction is regarded as positive, while the current flowing out of the junction is regarded as negative. In Fig. 6.4, O is a junction. According to sign convention, I_1, I_2, I_5 are positive and I_3, I_4, I_6 are negative. Then,

$$I_1 + I_2 - I_3 - I_4 + I_5 - I_6 = 0$$

$$\text{i.e., } I_1 + I_2 + I_5 = I_3 + I_4 + I_6$$

Thus the sum of currents entering the point O is equal to the sum of currents leaving it. In other words Kirchhoff's first law states, "When steady current flows in an electric circuit, there is no accumulation of charge at any point or junction."

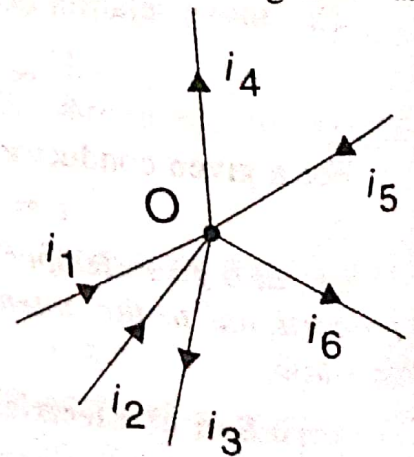


Fig. 6.4

Law 2. *The algebraic sum of the products of the current and resistance in any closed loop of a circuit is equal to the algebraic sum of electromotive forces (e.m.f.'s) acting in that loop. i.e., $\sum IR = \sum E$.*

Sign convention. (i) A product of current and resistance is taken as positive when we traverse in the direction of the current.

(ii) The emf is taken as positive when we traverse from the negative to the positive electrode of the cell through the electrolyte.

Let i_1 and i_2 be the currents through R_1 and R_2 (Fig. 6.5). Applying first law to the junction A, the current through R is $i_1 + i_2$. Then applying the second law for the closed loop ZAYZ,

$$i_1 R_1 + (i_1 + i_2) R = E_1$$

Similarly, for the closed loop XAYX,

$$i_2 R_2 + (i_1 + i_2) R = E_2$$

For the closed loop ZAXYZ,

$$i_1 R_1 - i_2 R_2 = E_1 - E_2$$

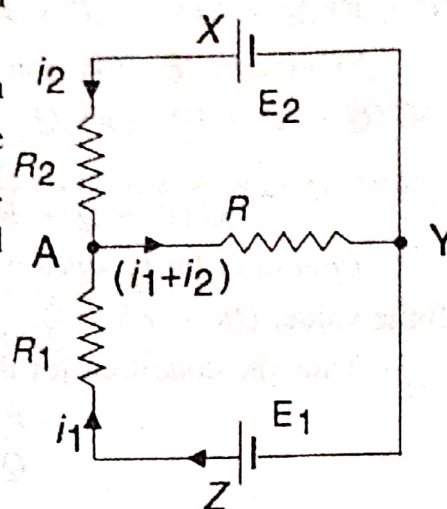


Fig. 6.5

These 'Loop equations' enable us to obtain the values of currents in different parts of the circuit in terms of R , R_1 , R_2 , E_1 and E_2 .

Four resistances P , Q , S and R form a closed network $ABCD$ (Fig. 6.6). A cell of emf E is connected between A and C . A galvanometer of resistance G is connected between B and D .

Let I be the current along EA , I_1 along AB and I_g along BD . By Kirchhoff's first law, the current along the various branches will be as shown.

We can obtain an expression for the current through the galvanometer i_g by applying Kirchhoff's voltage law (KVL).

Applying KVL to the closed mesh $ABDA$,

$$I_1 P + I_g G - (I - I_1) R = 0$$

$$\text{or } (P + R) I_1 + G I_g = I R \quad \dots (1)$$

Applying KVL to the closed mesh $BCDB$,

$$(I_1 - I_g) Q - (I - I_1 + I_g) S - I_g G = 0$$

$$\text{or } (Q + S) I_1 - (Q + S + G) I_g = I S \quad \dots (2)$$

Multiplying Eq (1) by $(Q + S)$,

$$(P + R) (Q + S) I_1 + G (Q + S) I_g = R (Q + S) I \quad \dots (3)$$

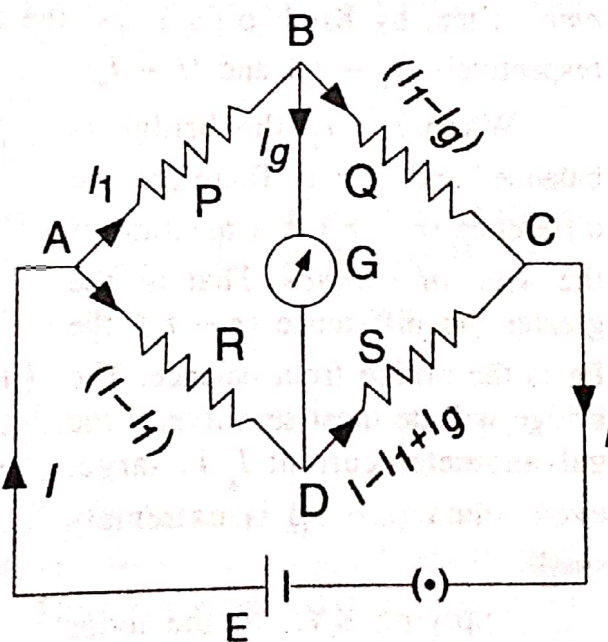


Fig. 6.6

Multiplying Eq. (2) by $(P + R)$,

$$(P + R)(Q + S)I_1 - (P + R)(Q + S + G)I_g = S(P + R)I \quad \dots(4)$$

Subtracting Eq. (4) from Eq. (3),

$$[G(Q + S) + (P + R)(Q + S + G)]I_g = [R(Q + S) - S(P + R)]I$$

$$\therefore I_g = \frac{(QR - PS)I}{[G(P + Q + R + S) + (P + R)(Q + S)]} \quad \dots(5)$$

Condition for balance is $i_g = 0$ i.e., $(QR - PS)I = 0$. Since I has a finite value, $QR - PS = 0$.

Thus the condition for the balance of the bridge is

$$\frac{P}{Q} = \frac{R}{S} \quad \dots(6)$$

Sensitivity of Wheatstone's bridge

Fig. 6.7 represents a Wheatstone's bridge $ABCD$. The unknown resistance r is connected in the arm CD . Let the resistances in the arms AB , BC and AD be nmr_1 , mr_1 and nr_1 respectively. Let G be the galvanometer resistance. Let I be the current through the unknown resistance r , I_g the current through the galvanometer and I_1 the current through resistance nmr_1 . Then, by Kirchhoff's I law, the currents through mr_1 and nr_1 are respectively $(I_1 + I_g)$ and $(I + I_g)$.

When $r = r_1$, the bridge is balanced and $I_g = 0$. Therefore, the difference $(r - r_1)$ is a measure of the *want of balance*. That is, the greater the difference $(r - r_1)$, the far is the bridge from balance. The bridge will be most sensitive, if the galvanometer current I_g is large, even when $(r - r_1)$ is extremely small.

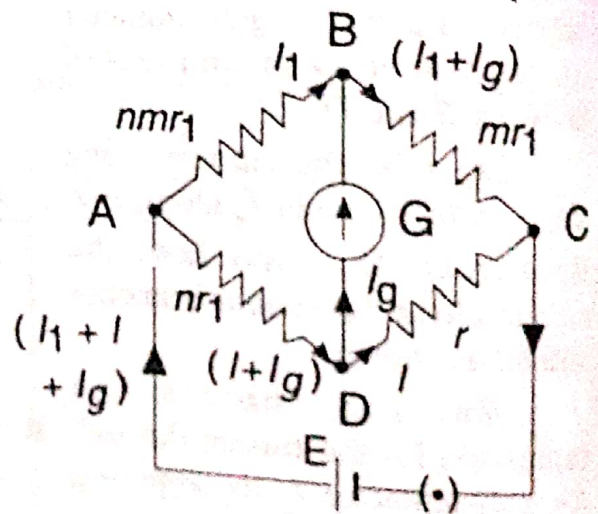


Fig. 6.7

Applying KVL to the loops $ABDA$ and $BCDB$ respectively,

$$nmr_1 I_1 - G I_g - nr_1 (I + I_g) = 0 \quad \dots(1)$$

$$mr_1 (I_1 + I_g) - rI + G I_g = 0 \quad \dots(2)$$

Multiplying Eq. (2) by n , we get

$$nmr_1 I_1 + nmr_1 I_g - nrI + nG I_g = 0 \quad \dots(3)$$

Subtracting Eq. (1) from Eq. (3), we get

$$G(1 + n)I_g + nr_1(1 + m)I_g - n(r - r_1)I = 0$$

$$\text{or } \frac{I_g}{I} = \frac{n(r - r_1)}{G(1 + n) + nr_1(1 + m)}$$

$$\text{or } \frac{I_g}{I} = \frac{r - r_1}{(1 + 1/n)G + r_1(1 + m)} \quad \dots(4)$$

For a given want of balance $(r - r_1)$, the value of I_g will be large when.

- (i) I , the current through the unknown resistance is large.
- (ii) G , the galvanometer resistance is small.
- (iii) n , the ratio $\frac{P}{Q} = \frac{nmr_1}{mr_1}$ is large.
- (iv) m , the ratio $\frac{P}{R} = \frac{nmr_1}{nr_1}$ is small.

The current I cannot be increased indefinitely because this will heat up the resistance coils and change their resistances. The sensitivity of the galvanometer itself is decreased if the resistance of the galvanometer G is too low.

Therefore, we have to increase the sensitivity by increasing n and decreasing m . In the ideal case, if $n = \infty$ and $m = 0$,

$$\left(\frac{I_g}{I} \right)_{\max} = \frac{(r - r_1)}{(G + r_1)}$$

But this maximum sensitivity can never be realised in practice, because when $n = \infty$, the resistances in AB and AD are infinite. If $m = 0$, the points A and C are short circuited.

If $m = n = 1$,

$$\frac{I_g}{I} = \frac{(r - r_1)}{2(G + r_1)}$$

This value is still half of the value in the ideal case. In practice, this is the condition for maximum sensitivity. However, for good sensitiveness of the bridge, $n > 1$ and $m < 1$. Thus the Callendar rule is : *The sensitivity of the bridge will be higher if the resistance in series with the unknown resistance is greater than the resistance connected in parallel to it.*

The positions of the galvanometer and the battery can be interchanged but these two arrangements are not equally sensitive. Maxwell gave the following rule regarding the best arrangement of these.

Out of the battery and the galvanometer, the one having the higher resistance should be connected between the junction of the two highest resistances and the junction of the two lowest resistances.

1. A battery of e.m.f. 6 volts and internal resistance 5 ohm is joined in parallel with another of e.m.f. 10 volts and internal resistance 1 ohm and the combination sends a current through an external resistance 12 ohms. Calculate the current through each battery.

Sol. Applying Kirchhoff's II law to the mesh ABFEA (Fig. 6.8),

$$I_1 \times 5 + (I_1 + I_2) \times 12 = 6$$

$$\text{or } 17I_1 + 12I_2 = 6 \quad \dots(1) \quad A$$

Applying Kirchhoff's II law to the mesh CDEFC

$$I_2 \times 1 + (I_1 + I_2) \times 12 = 10$$

$$\text{or } 12I_1 + 13I_2 = 10 \quad \dots(2) \quad E$$

Solving (1) and (2), we get

$$I_1 = -\frac{6}{11} \text{ A}, \quad I_2 = \frac{14}{11} \text{ A}.$$

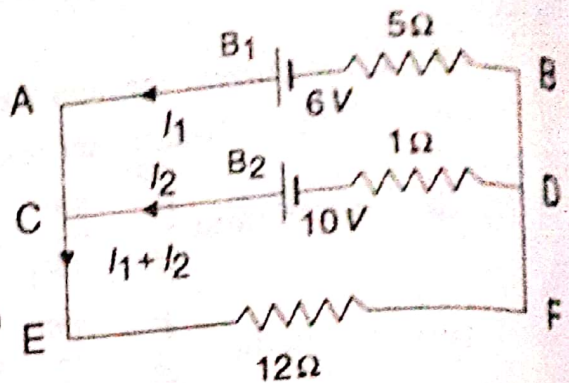


Fig. 6.8

2. Twelve conductors each of resistance 3 ohm are connected to form a skeleton cube. A battery of e.m.f. 6 V and internal resistance 0.5 ohm is connected between two diagonally opposite corners of the cube. Find the equivalent resistance of the cube, and the current supplied by the battery.

Sol. Let I be the current supplied by the battery. The current I enters the system at A and leaves it at G (Fig. 6.9). The distribution of current in the various branches is as shown in Fig. 6.9.

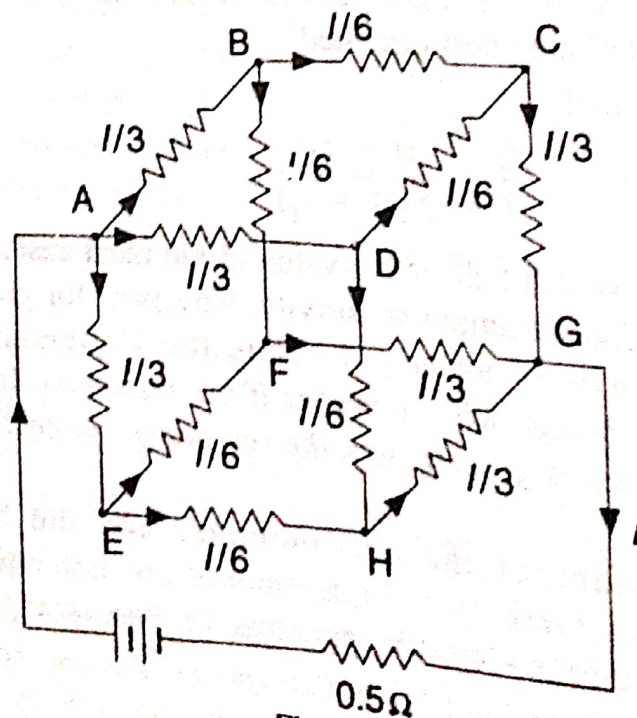


Fig. 6.9

Applying KVL to the closed loop ABFGA,

$$\frac{I}{3} \times 3 + \frac{I}{6} \times 3 + \frac{I}{3} \times 3 + 0.5I = 6$$

$$I + 0.5I + I + 0.5I = 6 \quad \text{i.e., } 3I = 6$$

$$\therefore I = 2 \text{ A}$$

Let X be the equivalent resistance of the frame work. Then,

$$(X + 0.5)I = 6$$

Substituting for I , $2X + 1 = 6$ or $2X = 5$

$$\therefore X = 2.5 \Omega$$

Example. The resistance of the four arms of a Wheatstone's net work are 5, 5, 5 and 5.2 ohms. The resistance of the galvanometer is 40 ohms and the battery supplies a current of 0.2 Amp. Calculate the current through the galvanometer.

Sol. Applying Kirchhoff's second law to the closed network ABDA (Fig. 6.10),

$$5I + 40I_g - 5[0.2 - I] = 0$$

$$\text{or } 10I + 40I_g = 1 \quad \dots(1)$$

For the network BDCB,

$$40I_g + 5.2(0.2 - I + I_g) - 5(I - I_g) = 0$$

$$50.2I_g - 10.2I = -1.04 \quad \dots(2)$$

$$\therefore I_g = -\frac{0.2}{910} = -0.0002198 \text{ A.}$$

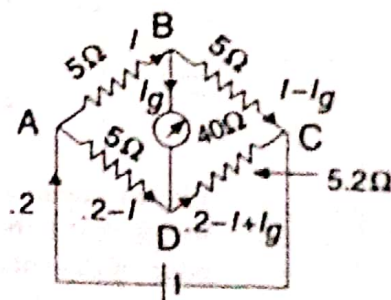


Fig. 6.10

EXERCISE VI

1. Define electric current and current density. Establish the relation between the current density and the velocity of the charge carriers.
2. Prove the equation of continuity.

$$\text{div } \mathbf{J} + \frac{d\rho}{dt} = 0.$$

3. Derive an expression for the electrical conductivity of a metal on Lorentz-Drude theory and obtain from it Ohm's law. How does conductivity vary with temperature according to above theory? How far is this result correct? What is Wiedemann and Franz law?
4. State Kirchhoff's laws of distribution of currents in an electrical network. Apply these laws to deduce the condition of balance of a Wheatstone's bridge.
5. Using Kirchhoff's laws, derive an expression for the sensitivity of Wheatstone's bridge. How do the positions of cell and galvanometer affect the sensitivity?
6. A current of 300 A exists in a copper wire whose diameter is 1 cm. Find the current density and calculate the electron drift speed. Given

Example 1. An e.m.f. 10 volts is applied to a circuit having a resistance of 10 ohms and an inductance of 0.5 henry. Find the time required by the current to attain 63.2% of its final value. What is the time constant of the circuit ?

Sol. $I = I_0 (1 - e^{-Rt/L})$

Given $\frac{I}{I_0} = \frac{63.2}{100}, \frac{R}{L} = \frac{10}{0.5} = 20$

$$\frac{63.2}{100} = 1 - e^{-20t}$$

$$e^{-20t} = 1 - 0.632 = 0.368$$

$$e^{20t} = \frac{1}{0.368} = 2.717$$

$$20t = \log_e 2.717$$

$$t = \frac{1}{20} \times 2.3026 \times \log_{10} 2.717$$

$$= \frac{2.3026 \times 0.4341}{20} = 0.05 \text{ sec.}$$

The time constant of the circuit is

$$\frac{L}{R} = \frac{0.5}{10} = \frac{1}{20} \text{ sec.}$$

Example 2. An inductance of 500 mH and a resistance of 5 ohms are connected in series with an e.m.f of 10 volts. Find the final current. If now the cell is removed and the two terminals are connected together, find the current after (i) 0.05 sec. and (ii) 0.2 sec.

Sol. Final current $I_0 = \frac{E}{R} = \frac{10}{5} = 2A$

During discharge, $I = I_0 e^{-Rt/L}$

Now $\frac{R}{L} = \frac{5}{500 \times 10^{-3}} = 10$

(i) When $t = 0.05 \text{ sec.}, I = 2e^{-10 \times 0.05} = 1.213A$

(ii) When $t = 0.2 \text{ sec.}, I = 2e^{-10 \times 0.2} = 0.271A$

12.3.

(a) A capacitor C and a resistance R are connected to a cell of emf E through a Morse key K (Fig. 12.6). When the key is pressed, a momentary current I flows through R . At any instant t , let Q be the charge on the capacitor of capacitance C .

P.D. across capacitor $= Q/C$

P.D. across resistor $= RI$

The emf equation of the circuit is

$$E = (Q/C) + RI \quad \dots (1)$$

$$E = (Q/C) + R(dQ/dt)$$

$$I = dQ/dt$$

The capacitor continues getting charged till it attains the maximum charge Q_0 . At that instant $I = dQ/dt = 0$. The P.D. across the capacitor is $E = Q_0/C$.

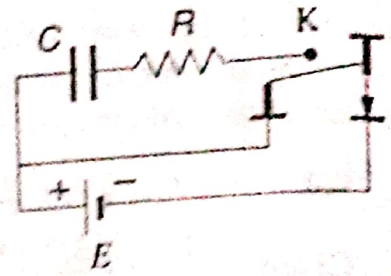


Fig. 12.6

i.e., when, $Q = Q_0, \frac{dQ}{dt} = 0$ and $E = \frac{Q_0}{C}$

$$\therefore \frac{Q_0}{C} = \frac{Q}{C} + R \frac{dQ}{dt}$$

$$(Q_0 - Q) = CR \frac{dQ}{dt}$$

$$\left(\frac{dQ}{Q_0 - Q} \right) = \frac{dt}{CR} \quad \dots (2)$$

Integrating, $-\log_e (Q_0 - Q) = \frac{t}{CR} + K$

where K is a constant.

When $t = 0, Q = 0 \therefore -\log_e Q_0 = K$

$$\therefore -\log_e (Q_0 - Q) = \frac{t}{CR} - \log_e Q_0$$

$$\log_e (Q_0 - Q) = -\frac{t}{CR} + \log_e Q_0$$

$$\log_e (Q_0 - Q) - \log_e Q_0 = -\frac{t}{CR}$$

$$\log_e \left(\frac{Q_0 - Q}{Q_0} \right) = -\frac{t}{CR}$$

$$\frac{Q_0 - Q}{Q_0} = e^{-\frac{t}{CR}} \quad \text{or} \quad 1 - \frac{Q}{Q_0} = e^{-\frac{t}{CR}}$$

$$\therefore Q = Q_0 \left(1 - e^{-\frac{t}{CR}} \right) \quad \dots (3)$$

The term CR is called time constant of the circuit.

At the end of time $t = CR, Q = Q_0 (1 - e^{-1}) = 0.632Q_0$

Thus, the time constant may be defined as the time taken by the capacitor to get charged to 0.632 times its maximum value.

The growth of charge is shown in Fig. 12.7.

The rate of growth of charge is

$$\frac{dQ}{dt} = \frac{Q_0}{CR} e^{-\frac{t}{CR}} = \frac{1}{CR} (Q_0 - Q)$$

Thus it is seen that smaller the product CR , the more rapidly does the charge grow on the capacitor.

The rate of growth of the charge is rapid in the beginning and it becomes less and less as the charge approaches nearer and nearer the steady value.

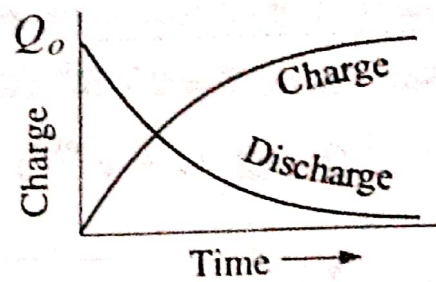


Fig. 12.7

(b) Discharging of a Capacitor through Resistance

Let the capacitor having charge Q_0 be now discharged by releasing the Morse key K (Fig. 12.6). The charge flows out of the capacitor and this constitutes a current. In this case $E = 0$.

$$R \frac{dQ}{dt} + \frac{Q}{C} = 0 \quad \dots (1)$$

or
$$\frac{dQ}{Q} = -\frac{1}{CR} dt$$

Integrating, $\log_e Q = -\frac{t}{CR} + K$, where K is a constant

When $t = 0$, $Q = Q_0 \therefore \log_e Q_0 = K$

$$\log_e Q = -\frac{t}{CR} + \log_e Q_0$$

or
$$\log_e \frac{Q}{Q_0} = -\frac{t}{CR} \quad \text{or} \quad \frac{Q}{Q_0} = e^{-\frac{t}{CR}}$$

$\therefore Q = Q_0 e^{-\frac{t}{CR}} \quad \dots (2)$

This shows that the charge in the capacitor decays exponentially and becomes zero after infinite interval of time (Fig. 12.7).

The rate of discharge is

$$I = \frac{dQ}{dt} = -\frac{Q_0}{CR} e^{-\frac{t}{CR}} = -\frac{Q}{CR} \quad \dots (3)$$

Thus, smaller the time-constant CR , the quicker is the discharge of the capacitor.

In Eq. (2), if we put $t = CR$, then $Q = Q_0 e^{-1} = 0.368 Q_0$.

Hence time constant may also be defined as the time taken by the current to fall from maximum to 0.368 of its maximum value.

1. A capacitor is charged by DC supply through a resistance of 2 megaohms. If it takes 0.5 seconds for the charge to reach three quarters of its final value, what is the capacitance of the capacitor?

Sol. Here, $R = 2 \times 10^6 \Omega$, $t = 0.5s$, $Q/Q_0 = 3/4$.

$$Q = Q_0 (1 - e^{-t/CR}) \quad \text{or} \quad \frac{Q}{Q_0} = (1 - e^{-t/CR})$$

$$\text{or} \quad \frac{3}{4} = \left(1 - e^{-\frac{0.5}{C \times (2 \times 10^6)}} \right)$$

$$\text{or} \quad e^{\frac{0.5}{C \times (2 \times 10^6)}} = 4 \quad \text{or} \quad \frac{0.5}{C \times (2 \times 10^6)} = \log_e 4$$

$$\text{or} \quad C = \frac{0.5}{(2.3026 \log_{10} 4) (2 \times 10^6)} = 0.18 \times 10^{-6} \text{ F} = 0.18 \mu\text{F}.$$

2. A capacitor of capacitance $0.1 \mu\text{F}$ is first charged and then discharged through a resistance of 10 megaohm. Find the time, the potential will take to fall to half its original value.

$$\text{Sol.} \quad Q = Q_0 e^{-t/CR} \quad \text{or} \quad \frac{Q}{Q_0} = e^{-t/CR} \quad \text{or} \quad \ln \left(\frac{Q_0}{Q} \right) = \frac{t}{CR}$$

$$\therefore t = CR \ln \left(\frac{Q_0}{Q} \right)$$

$$Q = CV \text{ and } Q_0 = CV_0. \quad \therefore \frac{Q_0}{Q} = \frac{V_0}{V}$$

$$\therefore t = CR \ln \left(\frac{V_0}{V} \right)$$

$$\text{Here, } C = 10^{-7} \text{ F; } R = 10^7 \Omega, V_0/V = 2.$$

$$\therefore t = 10^{-7} \times 10^7 \times \ln 2 = 0.6931 \text{ s}.$$

3. A resistance R and a $2 \mu\text{F}$ capacitor in series are connected to a 200 volt direct supply. Across the capacitor is a neon lamp that strikes at 120 volts. Calculate the value of R to make the lamp strike 5 seconds after switch has been closed.

Sol. The resistance R must be such that the p.d. across the capacitor should rise to 120 volt in 5 seconds after the switch is closed. The lamp would then strike.

The equation of charging is

$$Q = Q_0 (1 - e^{-t/RC})$$

$$\text{or } CV = CV_0 (1 - e^{-t/RC})$$

$$\text{or } V = V_0 (1 - e^{-t/RC})$$

Here, $V = 120$ volt, $V_0 = 200$ volt, $t = 5$ sec and

$$C = 2 \times 10^{-6} \text{ farad,}$$

$$\therefore 120 = 200 (1 - e^{-5/2 \times 10^{-6} R})$$

$$\text{or } e^{5/2 \times 10^{-6} R} = \frac{5}{2}$$

$$\frac{5}{2 \times 10^{-6} R} = \log_e 5 - \log_e 2 = 1.6094 - 0.6931 = 0.9163$$

$$R = \frac{5}{2 \times 10^{-6} \times 0.9163} = 2.73 \times 10^6 \text{ ohm} = 2.73 \text{ megaohm}$$

12.4. Measurement of High Resistance by Leakage

When a capacitor of capacitance C and initial charge Q_0 is allowed to discharge through a resistance R for a time t , the charge remaining on the capacitor is given by

$$Q = Q_0 e^{-t/CR}$$

$$\frac{Q_0}{Q} = e^{t/CR}$$

$$\log_e \frac{Q_0}{Q} = \frac{t}{CR}$$

$$\therefore R = \frac{t}{C \log_e (Q_0/Q)} = \frac{t}{2.3026 C \log_{10} (Q_0/Q)}$$

If R is high, CR will be high and the rate of discharge of capacitor will be very slow. Thus if we determine Q_0/Q from experiment, then R can be calculated.

Connections are made as shown in Fig. 12.8. C is a capacitor of known capacitance, R is the high resistance to be measured, B.G. is a ballistic galvanometer, E is a cell, and K_1 , K_2 , K_3 are tap keys.

Keeping K_2 and K_3 open, the capacitor is charged by depressing the key K_1 . K_1 is then opened and at once K_3 is closed. The capacitor discharges through

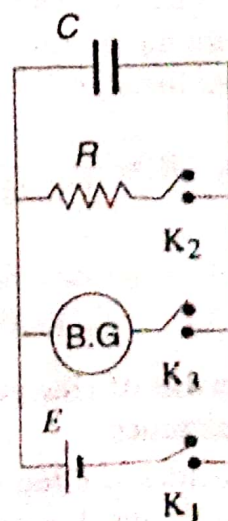


Fig. 12.8

the galvanometer which records a throw θ_0 . The throw θ_0 is proportional to Q_0 . The capacitor is again charged to the maximum value keeping K_2 and K_3 open and closing K_1 . K_1 is then opened and K_2 is closed for a known time t . Some of the charge leaks through R . K_2 is opened and at once K_3 is closed. The charge Q remaining on the capacitor then discharges through the galvanometer. The resulting throw θ is noted. Then $Q \propto \theta$

$$\text{Now, } \frac{Q_0}{Q} = \frac{\theta_0}{\theta}$$

$$\therefore R = \frac{t}{2.3026 C \log_{10} (\theta_0/\theta)}$$

A series of values of t and θ are obtained. A graph is plotted between t and $\log_{10} (\theta_0/\theta)$ which is a straight line. Its slope gives the mean value of $\frac{t}{\log_{10} (\theta_0/\theta)}$. As C is known, the value of R can be calculated.

Example 1. If the charge on a capacitor of capacitance $2 \mu\text{F}$ in leaking through a high resistance of 100 megaohms is reduced to half its maximum value, calculate the time of leakage.

$$\text{Sol. } R = \frac{t}{2.3026 C \log_{10} (Q_0/Q)}$$

$$\text{Here, } C = 2 \times 10^{-6} \text{ F, } R = 10^8 \Omega, Q_0/Q = 2.$$

$$\therefore t = 2.3026 CR \log_{10} (Q_0/Q) = 2.3026 (2 \times 10^{-6}) 10^8 \log_{10} 2 \\ = 138.7 \text{ s.}$$

Example 2. In an experiment to determine high resistance by leakage, a capacitor of $0.2 \mu\text{F}$ is used. It is first fully charged and discharged through a B.G. The observed kick was 12 cm on the scale. The capacitor was fully charged again and allowed to leak through R for 2 sec . The remaining charge in C gave a kick of 6 cm on the same scale when discharged through the B.G. Calculate R .

$$\text{Sol. } R = \frac{t}{2.3026 C \log_{10} (\theta_0/\theta)}$$

$$= \frac{2}{2.3026 \times (0.2 \times 10^{-6}) \log_{10} (12/6)} = 14.43 \text{ megaohms.}$$

12.5. Growth of charge in a Circuit with Inductance, Capacitance and Resistance

Consider a circuit containing an inductance L , capacitance C and resistance R joined in series to a cell of emf E (Fig. 12.9). When the key K is pressed, the capacitor is charged. Let Q be the charge on the capacitor and I the current in the circuit at an instant t during charging. Then, the p.d.

Alternating Current

13.1. I

Consider a rectangular coil of N turns and of length a and width b rotating with uniform angular velocity ω about its axis in a uniform magnetic field B (Fig. 13.1).

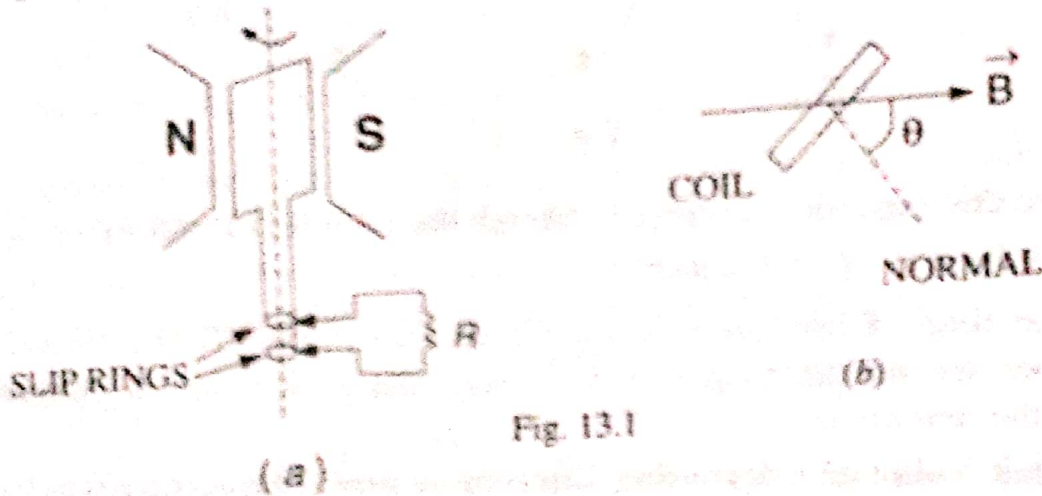


Fig. 13.1

The axis of rotation is at right angles to the field. As the coil rotates, the magnetic flux passing through it changes. Hence an emf is induced in the coil.

Suppose we start timing from the instant when the plane of the coil is at right angles to the field B , i.e., when the angle between the normal to the plane of the coil and the direction of the field is zero. Then, at an instant t , the normal to the plane of the coil will make angle $\theta (= \omega t)$ with the direction of B .

The magnetic flux linked with N turns of the coil is

$$\phi = NBA \cos \theta = NBA \cos \omega t.$$

where $A (= ab)$ is the area of the coil.

The instantaneous induced emf,

$$e = - \frac{d\phi}{dt} = - \frac{d}{dt} (NBA \cos \omega t) = NBA \omega \sin \omega t = E_0 \sin \omega t$$

Here, $E_0 = NBA\omega$, called the *peak value* of the e.m.f.

Now, $\omega = 2\pi\nu$ where ν = frequency of alternating voltage.

Thus when $\omega t = 0$, $\sin \omega t = 0$ and $E = 0$

$\omega t = \pi/2$, $\sin \omega t = 1$ and $E = E_0$

$\omega t = \pi$, $\sin \omega t = 0$ and $E = 0$

$\omega t = 3\pi/2$, $\sin \omega t = -1$ and $E = -E_0$

and $\omega t = 2\pi$, $\sin \omega t = 0$ and $E = 0$ again.

A graph of E against ωt is a sine curve (Fig. 13.2). Such an e.m.f. is called an 'alternating e.m.f.' The resulting current in the coil, if the coil is part of a closed circuit, is the 'alternating current.'

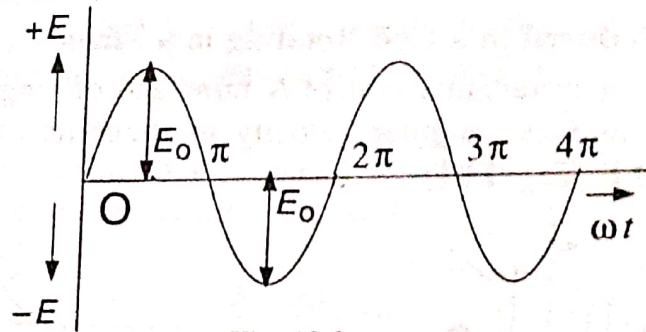


Fig. 13.2.

The corresponding current I through the circuit is given by

The time of one cycle is known as *time period* T , the number of cycles per second the *frequency* ν ($= 1/T$), the peak value of current or voltage the *amplitude*,

Peak value of alternating current or emf. The maximum value of alternating current or emf in the positive or negative direction is called *peak value of alternating current or emf*. It is denoted by I_0 or E_0 .

Mean value of alternating current is defined as its average over half a cycle.

$$\begin{aligned} I_{\text{mean}} &= \frac{\int_0^{T/2} I dt}{T/2} = \frac{\int_0^{\pi/\omega} I_0 \sin \omega t dt}{\pi/\omega} \\ &= \frac{I_0 \omega}{\pi} \left[\frac{-\cos \omega t}{\omega} \right]_0^{\pi/\omega} \\ &= -\frac{I_0}{\pi} [\cos \pi - \cos 0] \\ &= \frac{2I_0}{\pi} = 0.637 I_0 \end{aligned}$$

Similarly, $E_{\text{mean}} = 0.637 E_0$

It is defined as

the square root of the average of I^2 during a complete cycle.

$$\begin{aligned} I^2 &= \frac{\int_0^{2\pi/\omega} I^2 dt}{2\pi/\omega} = \frac{\int_0^{2\pi/\omega} I_0^2 \sin^2 \omega t dt}{2\pi/\omega} \\ &= \frac{I_0^2 \omega}{2\pi} \int_0^{2\pi/\omega} \frac{1}{2} (1 - \cos 2\omega t) dt \\ &= \frac{I_0^2 \omega}{4\pi} \left[t - \frac{\sin 2\omega t}{2\omega} \right]_0^{2\pi/\omega} \\ &= \frac{I_0^2 \omega}{4\pi} \left[\frac{2\pi}{\omega} \right] = \frac{I_0^2}{2} \end{aligned}$$

$$I_{rms} = \sqrt{I^2} = \frac{I_0}{\sqrt{2}} = 0.707 I_0$$

$$\text{Similarly, } E_{rms} = \frac{E_0}{\sqrt{2}} = 0.707 E_0$$

Form factor. The form factor gives an indication of the wave shape of the alternating voltage or current. It is defined as the ratio of the virtual or r.m.s. value to the average value of alternating current or voltage. Thus in the case of a sinusoidal current (or voltage), form factor is,

$$\text{Form factor} = \frac{I_{r.m.s.}}{I_{mean}} = \frac{E_{r.m.s.}}{E_{mean}} = \frac{0.707 E_0}{0.637 E_0} = 1.11$$

R.m.s. value. The rms value of an alternating current can also be defined as that direct current which produces the same rate of heating in a given resistance. Therefore, the r.m.s. value of alternating current is also called as the 'effective' or the 'virtual' value of the current.

$$I_{virtual} = \frac{I_0}{\sqrt{2}} = I_{rms}$$

Suppose an alternating current of instantaneous value $I = I_0 \sin \omega t$ is flowing through a circuit of resistance R .

$$\left. \begin{array}{l} \text{Total quantity of heat produced} \\ \text{over the complete cycle} \end{array} \right\} = H = \int_0^T I^2 R dt \quad \dots (1)$$

Let I_v stand for the root mean square or virtual value of the current.

Then the heat produced in time T is given by

$$H = I_v^2 RT \quad \dots (2)$$

Comparing Eqs. (1) and (2),

$$I_v^2 RT = \int_0^T I^2 R dt \quad \text{or} \quad I_v^2 T = \int_0^T I^2 dt$$

$$\begin{aligned} \text{or} \quad I_v^2 T &= \int_0^T I_0^2 \sin^2 \omega t dt = \frac{I_0^2}{2} \int_0^T 2 \sin^2 \omega t dt \\ &= \frac{I_0^2}{2} \int_0^T (1 - \cos 2\omega t) dt = \frac{I_0^2}{2} \left[t - \frac{\sin 2\omega t}{2\omega} \right]_0^T \\ &= \frac{I_0^2 T}{2} \end{aligned}$$

$$\text{or} \quad I_v^2 = \frac{I_0^2}{2}$$

$$\text{or} \quad I_v = \frac{I_0}{\sqrt{2}} = 0.707 I_0$$

Similarly, the r.m.s. value of an alternating voltage can be defined as that direct voltage which produces the same rate of heating in a given resistance. The r.m.s. value of alternating voltage is also called as the 'effective' or the 'virtual' value of the voltage.

$$E_{\text{virtual}} = \frac{E_0}{\sqrt{2}} = E_{\text{rms}}$$

In any circuit the ratio of the effective voltage to the effective current is defined as the impedance Z of the circuit.

13.2. AC Circuit Containing Resistance, Inductance and Capacitance in series (Series Resonance Circuit)

Let an alternating emf $E = E_0 \sin \omega t$ be applied to a circuit containing a resistance R , inductance L and capacitance C in series (Fig. 13.3). Let at any instant, I be the current in the circuit and Q be the charge on the capacitor.

The potential drop across the resistance $= RI$

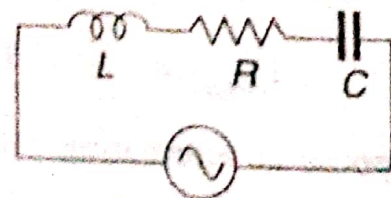
The E.M.F. induced in the

inductance $= L \frac{dI}{dt}$

The potential across the plates of the capacitor $= Q/C$.

$$\therefore L \frac{dI}{dt} + RI + \frac{Q}{C} = E_0 \sin \omega t$$

Differentiating with respect to t ,



$$E = E_0 \sin \omega t$$

Fig. 13.3

may be delta connected only when the load on the phases is the same or almost the same. Otherwise the machinery may be damaged by strong currents in the closed circuit of the windings.

14.3.

Principle. An induced emf is produced in a conductor whenever there is a change in the magnetic flux linked with it. It is a device which converts mechanical kinetic energy of rotation into electrical energy.

Construction. It consists of a rectangular coil of wire ABCD wound on a soft iron core (Fig. 14.6). It is mounted on a shaft between the poles of a powerful magnet. It can be rotated about an axis perpendicular to the direction of the magnetic lines of force. The ends of the coil are connected to two metal rings R_1 and R_2 which rotate with the coil. Contact is maintained with the external circuit by means of two carbon brushes B_1 and B_2 which touch the rings.

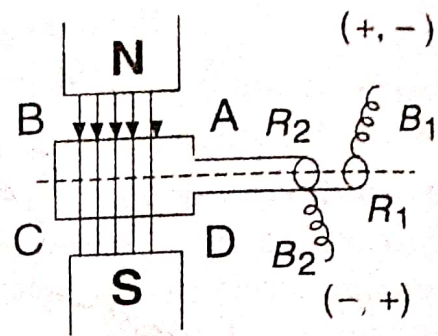


Fig. 14.6

Working. As the shaft rotates, magnetic flux through the coil changes. It produces an induced e.m.f in the coil which has an alternating nature. The e.m.f. sends a current through the external circuit completed through line wires and load. The current is also alternating. For this reason, this dynamo is also called alternator.

Theory. It has already been shown that the e.m.f at any instant is

$$E = E_0 \sin \omega t = BAN \omega \sin \omega t \text{ (See article 13.1)}$$

The generator described above is a *single phase generator*.

14.4. Two phase ac generator.

This generator consists of two sets of coils mutually at right angle to each other (Fig. 14.7). Each coil is provided with its own brushes and slip rings. The number of turns and area of cross section of each coil is the same. Hence the amplitude of emf generated in each coil will be the same.

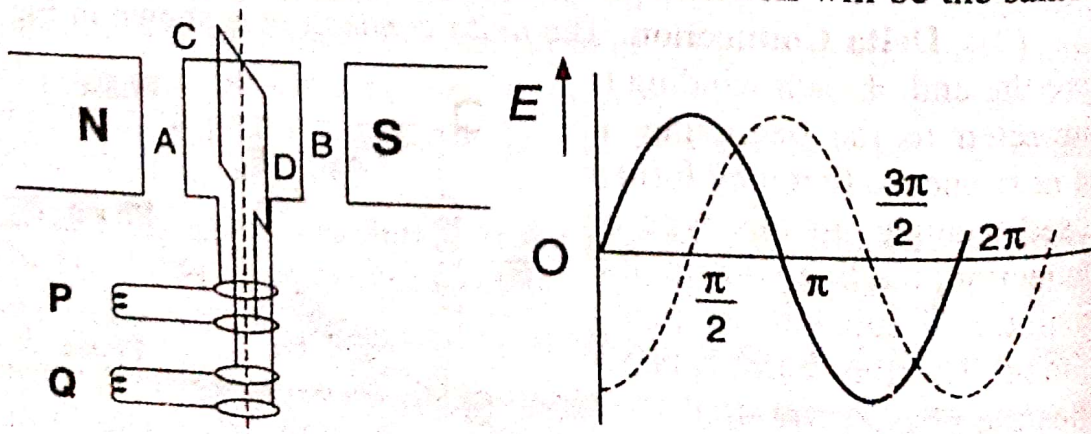


Fig. 14.7